Data Science 2024-2025

COMP5122M

Practical 2 (Week 4)

Lecturer: Duygu Sarikaya

Titanic: Machine Learning From Disaster Dataset

The sinking of the Titanic is one of the most infamous shipwrecks in history. On April 15, 1912, during her maiden voyage, the widely considered “unsinkable” RMS Titanic sank after colliding with an iceberg. Unfortunately, there weren’t enough lifeboats for everyone onboard, resulting in the death of 1502 out of 2224 passengers and crew. While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others. You will find the dataset (`train.csv`) that include passenger information like name, age, gender, class, etc.

In this assignment, you are asked some questions which will guide your exploratory data analysis of the dataset. You are not asked to answer the questions manually, please use your tools of choice for data analysis to answer the questions.

For this Practical, please work in pairs. You are asked to complete a report answering the questions below with a short explanation if the question asks for one.

If you prefer to use Python:

Please refer to the tutorial on Minerva to set up a running Python environment, Jupyter notebook, or simply log in on Google Colab, and import the libraries you will need. You can check the documentation of each library you will need such as NumPy, Pandas, Matplotlib (available online) to get more information about the functions you will use. Fort his assignment, you may also use visualization libraries such as seaborn or plotly.

If you prefer to use Tableau:

Please refer to the “Tableau getting started & tutorial” on Minerva.

Data Dictionary

Variable	Definition	Key
survival	Survival	0 = No, 1 = Yes
pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
sex	Gender	
Age	Age in years	

sibsp	# of siblings / spouses aboard the Titanic	
parch	# of parents / children aboard the Titanic	
ticket	Ticket number	
fare	Passenger fare	
cabin	Cabin number	
embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

If you are using Python:

For starters, you can open the csv (comma separated value) file and create a data frame using the pandas function read_csv:

import pandas as pd

df_titanic = pd.read_csv('../input/titanic/train.csv') // you should replace the path with your own

df_titanic.info() // shows information about the data frame you have just created for the Titanic dataset

If you are using Tableau, you can simply open the csv file or import it with Tableau CSV Connector.

Questions:

1. Please check how many missing values there are in the dataset for the **columns "Age","Cabin" and "Embarked"**. Missing values will have a null value (NaN). Do not forget to write how many missing values there are for each of these three columns in the comments.

   Important Note: For the rest of the assignment, you can delete the instances that have NaN values for **specific attributes (columns) asked in the related question.**

2. Please create a pie chart that shows the **percentage** of passengers that embarked from each port (Southampton, Cherbourg, Queenstown). Explain in your comments which port the most number of people embarked from and which port the least amount of people embarked from.

3. Please create a bar chart that shows the **number** of passengers traveling for each class. (You should have three bars referring to each class). Please explain which class had the most number of passengers.

4. Please create a plot that shows the **number** of **female passengers** for each ticket class who survived and who did not survive in **bar chart** format. (You should have three bar charts (for each ticket class) referring to passengers of ticket class 1 who survived and didn't survive, passengers of ticket class 2 who survived and didn't survive, and so on)

5. Please create **two boxplots** that show the **key age statistics of female and male passengers who survived and who did not survive.** (You should have two boxplots, one for female passengers and one for male passengers, each of these box plots having two boxes for the passengers who survived and who didn't, and each of these boxes should communicate key age statistics related to each group (e.g. female who survived)

6. Please create a cross table as shown below (X will be computed and included in your answer). The cross table makes it possible to get information about how many people for each gender have survived etc. Please indicate which gender has the most number of survivors. Explain which gender had a higher survival rate. What might be the reason?

Survived	**0**	**1**	**All**
Sex			
Female	X	X	X
Male	X	X	X
All	X	X	X

7. Please create a **heatmap** for the correlation between the attributes of survival, age, sex, fare, and ticket class.

8. Please calculate the **"Pearson" standard correlation coefficient** for each correlation mentioned in Q7 and show it on the heatmap you created for Q7. How is survival correlated to ticket class? Please explain.

9. Please plot the histograms that show the age distribution of the female and male passengers **who survived**. (You can arrange 10 bins for the range of ages [0 to 100].) You should have two histograms overlayed on top of each other in different colors. Please explain which gender had a higher survival chance by looking at these histograms.

10. Please plot a **scatterplot** that shows the **age–fare correlation** for **passengers who survived and who did not survive**. (You should have age on the x-axis and fare on the y-axis, and you should color code the passengers according to their survival).